Business Requirements Document (BRD)

AeroGrid

Low-Altitude Airspace Intelligence Warehouse

AeroGrid Development Team

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1 Document Purpose

This Business Requirements Document (BRD) details the stakeholder needs, operational business processes, analytical requirements, and operational constraints for the AeroGrid project.

2 Business Aims & Analytical Focus

The warehouse serves both day-to-day monitoring and long-term planning by addressing the following core needs:

- Delivering a unified operational awareness view for low-altitude airspace traffic.
- Quantifying airspace utilization across altitude layers and geographic grids.
- Detecting, locating, and analyzing close encounters (near-midair conflicts).
- Evaluating the operational impact of regulatory restrictions and adverse weather.
- Informing strategic expansion and corridor design for emerging drone operations.

3 Business Processes

The warehouse is structured around five core business processes representing actual operations in low-altitude airspace:

1. **Continuous Vehicle Tracking:** Captures ongoing position reports (position, height, speed, heading) broadcast by active aircraft and drones, forming the foundation for real-time awareness and density calculations.

2. **Complete Flight Journey Recording:** Records the entire path of each journey (takeoff to landing or monitored area entry/exit) to analyze route choices, duration, altitude profiles, and traffic flow.

3. **Restriction Activation and Coverage:** Records standing and temporary flight restrictions (TFRs around airports, stadiums, security sites), including effective time windows and geographic boundary polygons.

4. **Close Encounter Detection:** Identifies and logs instances where vehicles breach minimum safe separation distances in space and time, turning raw telemetry into measurable safety events.

5. **Airspace Occupancy Snapshot:** Calculates periodic utilization metrics (vehicle counts, occupancy duration per grid cell, busiest altitude bands) to support capacity planning.

4 Scope & Constraints

- **Data Scope:** Publicly available surveillance feeds (OpenSky), FAA UAS datasets, TartanAviation trajectories, and weather services.

- **Users:** Traffic management service providers, regulatory authorities, commercial drone operators, and municipal city planners.

- **Technical Constraints:** Designed using standard dimensional data warehousing principles (Star/Snowflake schemas) optimized for complex analytical queries.